

Defender Agent implementation

6/23/26 7:20 PM

Goal: Go thru my 3 defender agents come up w/ pseudocode & get ready to present approach & use AI to make a mini example code to better demonstrate

Patrol Defender: - What is it: Essentially I have 1 defender rather than static its dynamic ie I set two points and UUV would go back & forth

- Pseudocode:

Main idea ✓ 1) Blank map
↳ recreate the map visualizer tool in my own program
↳ just reuse & learn from Dons code

✓ 2) User picks points
↳ User select location & would need a variable x & y
↳ Select loc. B
↳ Safety check if "Object/something between" dont do it say error

✓ 3) Walk the Ocean?? Dumb name
↳ Prompt user if want to run A→B→A→B... infinite or nth times
↳ if user wanted A→B→C how to do cpp=vector store each in vector and after each point plotted

or could I set on timer ie every 30 sec move to next
or could it tell me time to circle back to original point
if user input = 'a' → place another → ask how fast do you want to travel to new ie (B→C)
else run program

Base code alone works for demo

- For actual implementation: Find out how to add agents alone both .h & .cpp
↳ Can I somehow have main code User picks UUV in Static or Dynamic

Terminal need to know dont fully set but where were at:

most used & keep needed
[cmake --build build -was used to build map
./build/Patrol Demo -need to do everytime?
rest?
[cd build
make
cd ..

Things I need to do: - Animate & understand code
- Understand how .h & .cpp files work to some extent
- could I make my own map - nuketown 2025?? water map